Task # 05:

Design a Java program to represent a College and its Students using a One-to-Many Has-A relationship. A college can have multiple students, but each student belongs to only one college. Use an array of Student objects to store the students.

First, create a Student class. It should store a student's ID, name, and marks. Provide a constructor to set these values when creating a Student object. Add a method called displayStudentDetails() to print the student's information, and a method isPassed() that returns true if the student's marks are 40 or above, otherwise false.

Next, create a College class. It should store the college name, city, and an array of Student objects. Provide a constructor to initialize these values. Add a method displayCollegeDetails() that prints the college information, displays details of all students, and calculates and shows the total number of students, the number who passed, the number who failed, and the average marks of all students.

In the main method, create at least five Student objects and store them in an array. Then create a College object using this array and call displayCollegeDetails() to display all the required information.